Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A source approaching a stationary listener at 15 m/s emits 500 Hz. Use v_{sound}=340 and $f_{obs}=f_s\ v/(v-v_s)$. Find observed frequency.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $1/f = 1/d_o + 1/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is approximately undeviated in the thin-lens model.

Your turn: Draw these two rays and mark their intersection for an object beyond 2 f.

Marzano check - circle: ① ② ③

03 Represent and interpret

Complete ray sketches for parallel rays through a converging lens and a diverging lens. Label the optical axis.



Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

04 Solve and justify

A converging lens has f=10 cm and object distance 30 cm. Find image distance and magnification $m=-d_i/d_o$.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A magnifying glass produces an upright virtual image when object is inside f. Explain why “converging lens always makes a real image”is wrong.

Marzano check - circle: 1 2 3

06 Choose a method

A wave snapshot has crests at 0.2, 0.6, 1 m. Find wavelength. Its period is 0.06 s; find frequency.

Marzano check - circle: 1 2 3

07 Exit check

For a converging lens f=10 cm and d_o=5 cm, find d_i and m. State image kind.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A lens has f=10 cm and object distance 30 cm. Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$. Find image distance and magnification.

Marzano check - circle: 1 2 3